66-week study module

Index — every file, what is in it, and the order to read it

34 files: 671 pages of PDF, a 155-problem tracker, an 11-sheet workbook and seven data files. Keep them all in one folder, including `problem-data`, and every link in the tracker works. Files are numbered in reading order, so a folder listing sorted by name is the reading order.

Start here

Open **study-tracker.html** in Chrome or Firefox. Its Today tab gives the session, the reading desk names the module file and the problems keyed to that week, and the Materials tab lists everything with one-click links. If a link does not open, read `setup-multi-device.pdf` — on an iPad they cannot work until the folder is hosted.

1 · The module, numbered in reading order

File	Weeks	Pages	Contents
module-01-foundations-wk01-10.pdf	1–10	35	Biology foundations for engineers — your original part 1
module-02-core-concepts-wk01-10.pdf	1–10	39	Core concepts for weeks 1–10, compiled: ten chapters, every number computed, 65-term glossary
module-03-framework-wk11-19.pdf	11–19	37	The framework: modelling, control, and the numbers that decide a design
module-04-building-the-strain-wk20-25.pdf	20–25	22	Building the strain: assembly, assays, genome engineering
module-05-designing-the-parts-wk26-31.pdf	26–31	26	Designing the parts: biosensors, regulation, libraries, protein design, natural products
module-06-reading-the-data-wk32-34.pdf	32–34	17	Reading the data: integration, sequencing, RNA-seq, the biofoundry
module-07-constraint-models-wk35-38.pdf	35–38	18	Constraint-based modelling: FBA, COBRApy, reconstruction, MATLAB
module-08-model-to-design-wk39-44.pdf	39–44	24	From model to design: strain design, machine learning, labelling, the bioreactor
module-09-product-recovery-wk45-48.pdf	45–48	20	Getting the product out: toxicity, in situ removal, two-phase modelling, immobilisation
module-10-trusting-numbers-wk49-52.pdf	49–52	19	Trusting the numbers: economics, validation, omics, host defences
module-11-product-classes-wk53-57.pdf	53–57	20	What the field has made: five product classes and their constraints
module-12-hosts-wk58-61.pdf	58–61	18	Hosts: what each organism gives you and what it charges
module-13-the-profession-wk62-66.pdf	62–66	20	The profession: publishing, writing, planning, the proposal, the application

2 · Extension modules, written after the 66 weeks

File	Pages	Contents
module-14-supplementary-chapters.pdf	24	Six supplementary chapters S1–S6: separations, biomass deconstruction, two-substrate kinetics, consortia, quantitative omics, control theory
module-15-protein-products.pdf	15	Protein products — secretion, inclusion bodies, refolding, glycosylation — plus five theory chapters, and problems PR1–PR8
module-16-multiphase-and-doe.pdf	13	Three-phase and solid-state fermentation, gel immobilisation as a process, the pathway-modification playbook, design of experiments as a method, and problems MP1–MP4 and DE1–DE4
module-17-editing-screening-lca.pdf	13	Cas12a, base and prime editing; flavonoids and P450s; FACS and droplets; life cycle analysis; and problems EV1–EV8
module-18-stability-and-screening-funnel.pdf	14	The stability chain vial-to-harvest, ALE campaign design, the screening funnel, contamination and phage, and problems SB1–SB4 and SC1–SC4
module-19-heterogeneity-and-dsp.pdf	9	Single-cell noise, bimodality and persisters wired into sorting; the upstream/downstream capital split for a commodity and a biologic; foam and debris quantified; problems HN1–HN3 and DS1–DS3

Each chapter in these says which weeks it extends, so they can be read when you reach the relevant week rather than at the end.

3 · Problem sets

File	Pages	Contents
problems-01-X1-X24-Y1-Y6.pdf	17	X1–X24 and Y1–Y6 — balances, kinetics, transport, statistics, data, probability, economics; plus six set on real public datasets you download yourself
problems-02-Z1-Z9.pdf	9	Z1–Z9 — downstream separations, quantitative omics, consortia
problems-03-MC1-MC5-R1-R5.pdf	8	MC1–MC5 on model curation, and R1–R5 worked on the supplied instrument data
problems-04-PP1-PP12.pdf	11	PP1–PP12 — the pipeline end to end, host choice through to a costed process
problems-05-corrigendum-and-verification.pdf	2	Corrigendum: the appendix holds 56 problems, not 54 — B4 and G2 were missed by a faulty extraction of mine. B4 verified, the G2 claim retracted, the verification restated for all 56

Modules 15 to 19 carry their own problems as well — PR1–PR8, MP1–MP4 and DE1–DE4, EV1–EV8, SB1–SB4 and SC1–SC4, HN1–HN3 and DS1–DS3. All 155 problems are also inside the tracker with statement, questions and worked solution.

4 · The tracker, your own documents, and the tools

File	Size	Contents
study-tracker.html	—	**THE TRACKER.** 66 weeks, 330 sessions, 155 problems with statement, questions and worked solution, a Materials tab linking every file, a competency tier map, week grid and search
study-tracker-original.html	—	The original tracker, untouched, as a fallback
00-complete-volume.pdf	461	Every PDF merged into one bookmarked volume, for reading on a tablet
curriculum-66-weeks-56-problems.pdf	210	Your curriculum, with the 56-problem appendix from its page 140
lookup-workbook.xlsx	11 sheets	Fillable workbook: the numbers to look up, live formulas, and sheet 10 listing every unverified claim
reading-list-by-tier.pdf	3	Books, papers and tools by competency tier, against module chapters and problems
reading-map-by-week.pdf	4	Textbook chapter per week, with a blank column for your own page numbers
setup-multi-device.pdf	2	How to host the folder so the links work on an iPad and a Mac
problem-data/ (7 files)	200 kB	Data for problems R1–R5, plus the answers file to open only after you have an answer

Cross-references: the old part numbers

The chapters were written as numbered "parts" and cross-reference each other that way. The files are now numbered in reading order instead, so this table maps between them.

File	Called in the text	Weeks	File	Called in the text	Weeks
Module 01	Part 01	1–10	Module 02	Part 14	1–10
Module 03	Part 02	11–19	Module 04	Part 07	20–25
Module 05	Part 08	26–31	Module 06	Part 09	32–34
Module 07	Part 10	35–38	Module 08	Part 11	39–44
Module 09	Part 03	45–48	Module 10	Part 04	49–52
Module 11	Part 05	53–57	Module 12	Part 06	58–61
Module 13	Part 12	62–66	Module 14	Part 13	supplementary
Problems 01	Part 15	—	Problems 02	Part 16	—
Problems 03	Part 17	—	Problems 04	Part 18	—
Module 15	Part 19	—	Module 16	Part 20	—
Module 17	Part 21	—	Module 18	Part 22	—
Module 19	Part 23	—			

Two things to do first

1. Read `problems-05-corrigendum-and-verification.pdf`. It records a counting error of mine and its correction: the appendix

holds 56 problems, not the 54 I reported for several rounds, because a text extraction of mine silently dropped two.

2. Open sheet 10 of the lookup workbook. It lists every claim in the module that came from memory rather than from a calculation, with where to verify each. All arithmetic was computed and re-checked; those inputs were not, because this was written without network access.